IEOR 221 - HW 5 - Jie Wen

1

Inputs: $S_0 = 100$, $K = 100$, $u = 1.1$, $d = 0.9$, $\Delta t = 0.5$ yr, $r = 8\%$ (cont.).
Risk-neutral up-probability $p = \dfrac{e^{r\Delta t} - d}{u - d} = \dfrac{e^{0.04} - 0.9}{0.2} \approx 0.7041$.

Stock lattice: $S_{uu} = 100 \cdot 1.1^2 = 121$, $S_{ud} = S_{du} = 99$, $S_{dd} = 81$.
Terminal call payoffs: $C_{uu} = \max(121 - 100, 0) = 21$, $C_{ud} = C_{dd} = 0$.

Backward induction with discount factor $e^{-r\Delta t} = e^{-0.04}$:
$C_u = e^{-0.04}(p \cdot 21 + (1-p) \cdot 0) \approx 14.2054$
$C_d = e^{-0.04}(p \cdot 0 + (1-p) \cdot 0) = 0$
$C_0 = e^{-0.04}(pC_u + (1-p)C_d) \approx 9.6092$ dollars.

So, the 1-year European call's fair value is **about $9.61**.

2

European put value:

Inputs: $S_0 = 100, K = 100, u = 1.1, d = 0.9, \Delta t = 0.5, r = 8\%$ (cont.).
Risk-neutral up-probability $p = \dfrac{e^{r\Delta t} - d}{u - d} \approx \dfrac{e^{0.04} - 0.9}{0.2} \approx 0.7041$; discount per half-year $e^{-r\Delta t} = e^{-0.04}$.

Terminal **put** payoffs at $T = 1$:
$S_{uu} = 121 \Rightarrow P_{uu} = 0, \quad S_{ud} = 99 \Rightarrow P_{ud} = 1, \quad S_{dd} = 81 \Rightarrow P_{dd} = 19.$
Step back one level:
$P_u = e^{-0.04}[p \cdot 0 + (1-p) \cdot 1] = e^{-0.04}(1-p) \approx 0.28434,$
$P_d = e^{-0.04}[p \cdot 1 + (1-p) \cdot 19] \approx 6.07894.$
Then at $t = 0$:
$P_0 = e^{-0.04}[pP_u + (1-p)P_d] \approx 1.9208.$

Put–call parity check

For European options (no dividends):
$C_0 - P_0 = S_0 - Ke^{-rT}$.
(Equivalently $C_0 + Ke^{-rT} = P_0 + S_0$.)

From the previous problem $C_0 \approx 9.6092$. Compute both sides:

- LHS = $9.6092 - 1.9208 = 7.6884$.
- RHS = $100 - 100e^{-0.08} \approx 100 - 92.3116 = 7.6884$.

They match (up to rounding), so **put–call parity holds**.

3

In CRR, $u = e^{\sigma\sqrt{\Delta t}}, d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}$ and the risk-neutral probability is $p = \dfrac{e^{r\Delta t} - d}{u - d}$.
Inputs: $S_0 = 140, \sigma = 25\% = 0.25, r = 4\% = 0.04, \Delta t = 0.25$

(a) Percentage up movement

$u = e^{0.25\sqrt{0.25}} \approx 1.133148 \Rightarrow$ **+13.3148%**.

(b) Percentage down movement

$d = e^{-0.25\sqrt{0.25}} \approx 0.882497 \Rightarrow$ **−11.7503%**.

(c) Risk-neutral probability of an up move

$p = \dfrac{e^{0.04 \cdot 0.25} - 0.882497}{1.133148 - 0.882497} \approx \mathbf{0.5088868}$.

(d) Risk-neutral probability of a down move

$1 - p \approx \mathbf{0.4911132}$.

Two-step valuation (6 months total)

Per step discount factor: $DF = e^{-r\Delta t} = e^{-0.01} \approx 0.9900498$. The price tree (recombining under CRR) and payoffs:

- Stock nodes at $T = 0.5$:
  $S_{uu} = S_0 u^2 \approx 179.7636, S_{ud} = S_0 \approx 140, S_{dd} = S_0 d^2 \approx 109.0321$.

European call (K = 150)

Terminal payoffs:
$C_{uu} = \max(179.7636 - 150, 0) = 29.7636, C_{ud} = 0, C_{dd} = 0$.

Backward induction:
$C_u = DF[pC_{uu} + (1-p)C_{ud}] \approx 0.9900498 \times (0.5088868 \times 29.7636) = 15.0400$
$C_d = DF[pC_{ud} + (1-p)C_{dd}] = 0$
$C_0 = DF[pC_u + (1-p)C_d] \approx \mathbf{7.5551}$.

European put (K = 150)

Terminal payoffs:
$P_{uu} = 0, P_{ud} = 10, P_{dd} = 40.9679$.

Backward induction:
$P_u = DF[p \cdot 0 + (1-p) \cdot 10] \approx 4.8625$
$P_d = DF[p \cdot 10 + (1-p) \cdot 40.9679] \approx 19.7044$
$P_0 = DF[pP_u + (1-p)P_d] \approx \mathbf{14.5849}$.

Quick check (put–call parity, $T = 0.5$)

$C_0 - P_0 \approx 7.5551 - 14.5849 = -7.0298$ and
$S_0 - Ke^{-rT} = 140 - 150e^{-0.02} = -7.0298$ ✓.

4

Use the binomial hedging rule: $\Delta = \frac{f_u - f_d}{S_u - S_d}$ at each node (shares per option).
For CRR, $u = e^{\sigma\sqrt{\Delta t}}, d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}$ and the risk-neutral probability is $p = \frac{e^{r\Delta t} - d}{u - d}$.

Inputs: $S_0 = 140, \sigma = 0.25, r = 0.04, \Delta t = 0.25$ yr $\Rightarrow u \approx 1.133148, d \approx 0.882497$.

First 3-month period (at $t = 0$)

Compute one-step option values (risk-neutral backward step), then the initial delta:

- $C_u \approx 14.9956, C_d = 0$ (from the tree's next step),
- $S_u = 140u \approx 158.6408, S_d = 140d \approx 123.5496$.
- $\Delta_0 = \frac{C_u - C_d}{S_u - S_d} \approx 0.427331$ **shares per option.**

For **10,000 options** (writer hedges by **buying** Δ shares):

- **Hold** $\approx 0.427331 \times 10{,}000 = 4{,}273.3$ **shares** (≈ **4,273** shares if rounding to whole shares).

Second 3-month period (rebalance at $t = 3$ months)

1. **If the stock moved UP to S_u:**
   Terminal call payoffs: $C_{uu} > 0, C_{ud} = 0$.
   $\Delta_{\text{up}} = \frac{C_{uu} - C_{ud}}{S_{uu} - S_{ud}} = \frac{29.7636 - 0}{179.7636 - 140} \approx \mathbf{0.748513}$ shares/option.
   For 10,000 options: ≈ 7,485 **shares**.
2. **If the stock moved DOWN to S_d:**
   Terminal call payoffs: $C_{ud} = 0, C_{dd} = 0$.
   $\Delta_{\text{down}} = \frac{0 - 0}{140 - 109.0321} = \mathbf{0}$.
   For 10,000 options: **0 shares**.